

Claude Code + Quarry: A Z Specification

Layer 2 — Knowledge Capture State

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1 Introduction

This specification extends the base Claude Code state machine (`claude-code.tex`) with quarry automatic knowledge capture state. Quarry provides local semantic search over codebases, documents, and conversation history.

The key properties verified by this model:

1. **Knowledge before work:** the codebase is indexed before Claude begins processing. The `SessionStart` hook auto-registers and syncs the working directory.
2. **No knowledge loss on compaction:** before context compaction shrinks the conversation, the `PreCompact` hook captures session notes. No path reaches `CompactContext` without the capture flag being set.
3. **WebFetch auto-capture:** URLs fetched during a session are automatically ingested into the `web-captures` collection.
4. **Capture toggles are respected:** each auto-capture layer (session sync, web fetch, compaction) can be independently disabled via `.punt-labs/quarry/config.md`.

2 Base Types (from `claude-code.tex`)

[*TOOLNAME*, *TOOLINPUT*, *TOOLOUTPUT*, *PROMPT*,
RESPONSE, *AGENTID*, *SESSIONID*, *CONTEXTCHUNK*]

SessionPhase ::= *spInactive* | *spIdle* | *spProcessing* | *spResponding* | *spEnded*

ToolPhase ::= *tpNone* | *tpPreHook* | *tpPermission* | *tpExecuting* | *tpPostHook*

PermissionMode ::= *pmDefault* | *pmPlan* | *pmAcceptEdits* | *pmDontAsk* | *pmBypass*

PermissionDecision ::= *pdAllow* | *pdDeny* | *pdAsk*

HookDecision ::= *hdAllow* | *hdBlock*

SessionStartSource ::= *ssStartup* | *ssResume* | *ssClear* | *ssCompact*

SessionEndReason ::= *seClear* | *seLogout* | *sePromptExit* | *seBypassDisabled* | *seOther*

ZBOOL ::= *ztrue* | *zfalse*

3 Quarry Types

[*DBNAME*, *URL*]

DBNAME identifies a quarry database (e.g., `default`, `punt-labs`). *URL* identifies a fetched web resource.

4 Global Constants

<i>maxTools</i> : $\mathbb{N}$
<i>maxSubagents</i> : $\mathbb{N}$
<i>maxContext</i> : $\mathbb{N}$
<i>maxCaptures</i> : $\mathbb{N}$
<i>maxTools</i> = 5
<i>maxSubagents</i> = 2
<i>maxContext</i> = 3
<i>maxCaptures</i> = 5

5 State

State

sessionId : $\mathbb{F}$ *SESSIONID*
sessionPhase : *SessionPhase*
permMode : *PermissionMode*
toolPhase : *ToolPhase*
currentTool : $\mathbb{F}$ *TOOLNAME*
toolsThisTurn : $\mathbb{N}$
activeSubagents : $\mathbb{F}$ *AGENTID*
context : seq *CONTEXTCHUNK*
pendingPrompt : $\mathbb{F}$ *PROMPT*
pendingResponse : $\mathbb{F}$ *RESPONSE*
codebaseIndexed : *ZBOOL*
activeDb : $\mathbb{F}$ *DBNAME*
capturedUrls : $\mathbb{F}$ *URL*
compactionSaved : *ZBOOL*
cfgSessionSync : *ZBOOL*
cfgWebFetch : *ZBOOL*
cfgCompaction : *ZBOOL*

$\#sessionId \leq 1$
 $sessionPhase = spInactive \Rightarrow sessionId = \emptyset$
 $sessionPhase \neq spInactive \Rightarrow \#sessionId = 1$
 $sessionPhase \neq spProcessing \Rightarrow toolPhase = tpNone$
 $\#currentTool \leq 1$
 $toolPhase = tpNone \Rightarrow currentTool = \emptyset$
 $toolPhase \neq tpNone \Rightarrow \#currentTool = 1$
 $toolsThisTurn \leq maxTools$
 $sessionPhase = spInactive \Rightarrow activeSubagents = \emptyset$
 $sessionPhase = spEnded \Rightarrow activeSubagents = \emptyset$
 $\#activeSubagents \leq maxSubagents$
 $\#context \leq maxContext$
 $\#pendingPrompt \leq 1$
 $sessionPhase = spIdle \Rightarrow pendingPrompt = \emptyset$
 $sessionPhase = spInactive \Rightarrow pendingPrompt = \emptyset$
 $sessionPhase = spEnded \Rightarrow pendingPrompt = \emptyset$
 $\#pendingResponse \leq 1$
 $sessionPhase \neq spResponding \Rightarrow pendingResponse = \emptyset$
 $\#activeDb \leq 1$
 $\#capturedUrls \leq maxCaptures$
 $sessionPhase = spInactive \Rightarrow codebaseIndexed = zfalse$
 $sessionPhase = spInactive \Rightarrow activeDb = \emptyset$
 $sessionPhase = spInactive \Rightarrow capturedUrls = \emptyset$
 $sessionPhase = spInactive \Rightarrow compactionSaved = zfalse$

6 Initialization

Init
State'

sessionId' = $\emptyset$
sessionPhase' = *spInactive*
permMode' = *pmDefault*
toolPhase' = *tpNone*
currentTool' = $\emptyset$
toolsThisTurn' = 0
activeSubagents' = $\emptyset$
context' = $\langle \rangle$
pendingPrompt' = $\emptyset$
pendingResponse' = $\emptyset$
codebaseIndexed' = *zfalse*
activeDb' = $\emptyset$
capturedUrls' = $\emptyset$
compactionSaved' = *zfalse*
cfgSessionSync' = *ztrue*
cfgWebFetch' = *ztrue*
cfgCompaction' = *ztrue*

7 Quarry Frame Condition

QuarryFrame
 Δ *State*

codebaseIndexed' = *codebaseIndexed*
activeDb' = *activeDb*
capturedUrls' = *capturedUrls*
compactionSaved' = *compactionSaved*
cfgSessionSync' = *cfgSessionSync*
cfgWebFetch' = *cfgWebFetch*
cfgCompaction' = *cfgCompaction*

8 Session Lifecycle Operations

8.1 StartSessionWithSync

SessionStart hook fires. If *cfgSessionSync* = *ztrue*, quarry auto-registers the working directory and syncs. The codebase is indexed before Claude can process prompts.

StartSessionWithSync

Δ State

sid? : *SESSIONID*

source? : *SessionStartSource*

mode? : *PermissionMode*

sessionPhase = *spInactive*
cfgSessionSync = *ztrue*
sessionPhase' = *spIdle*
sessionId' = {*sid?*}
permMode' = *mode?*
toolPhase' = *tpNone*
currentTool' = $\emptyset$
toolsThisTurn' = 0
activeSubagents' = $\emptyset$
context' = $\langle \rangle$
pendingPrompt' = $\emptyset$
pendingResponse' = $\emptyset$
codebaseIndexed' = *ztrue*
activeDb' = $\emptyset$
capturedUrls' = $\emptyset$
compactionSaved' = *zfalse*
cfgSessionSync' = *cfgSessionSync*
cfgWebFetch' = *cfgWebFetch*
cfgCompaction' = *cfgCompaction*

8.2 StartSessionNoSync

Session starts but sync is disabled. The codebase is not indexed automatically.

StartSessionNoSync

Δ State

sid? : *SESSIONID*

source? : *SessionStartSource*

mode? : *PermissionMode*

sessionPhase = *spInactive*
cfgSessionSync = *zfalse*
sessionPhase' = *spIdle*
sessionId' = {*sid?*}
permMode' = *mode?*
toolPhase' = *tpNone*
currentTool' = $\emptyset$
toolsThisTurn' = 0
activeSubagents' = $\emptyset$
context' = $\langle \rangle$
pendingPrompt' = $\emptyset$
pendingResponse' = $\emptyset$
codebaseIndexed' = *zfalse*
activeDb' = $\emptyset$
capturedUrls' = $\emptyset$
compactionSaved' = *zfalse*
cfgSessionSync' = *cfgSessionSync*
cfgWebFetch' = *cfgWebFetch*
cfgCompaction' = *cfgCompaction*

8.3 SubmitPrompt

SubmitPrompt

QuarryFrame

prompt? : *PROMPT*

hookResult? : *HookDecision*

contextChunk? : *CONTEXTCHUNK*

sessionPhase = *spIdle*

hookResult? = *hdAllow*

sessionPhase' = *spProcessing*

pendingPrompt' = {*prompt?*}

context' = (1 .. *maxContext*) $\triangleleft$ (*context* $\hat{\ } \langle$ *contextChunk?* $\rangle$)

toolsThisTurn' = 0

toolPhase' = *tpNone*

currentTool' = $\emptyset$

sessionId' = *sessionId*

permMode' = *permMode*

activeSubagents' = *activeSubagents*

pendingResponse' = $\emptyset$

RejectPrompt

$\exists$ *State*

prompt? : *PROMPT*

hookResult? : *HookDecision*

sessionPhase = *spIdle*

hookResult? = *hdBlock*

9 Tool Call Operations

9.1 BeginToolCall

BeginToolCall

QuarryFrame

tool? : *TOOLNAME*

sessionPhase = *spProcessing*

toolPhase = *tpNone*

toolsThisTurn < *maxTools*

toolPhase' = *tpPreHook*

currentTool' = {*tool?*}

sessionPhase' = *sessionPhase*

sessionId' = *sessionId*

permMode' = *permMode*

toolsThisTurn' = *toolsThisTurn*

activeSubagents' = *activeSubagents*

context' = *context*

pendingPrompt' = *pendingPrompt*

pendingResponse' = *pendingResponse*

9.2 PreToolHookAllow

PreToolHookAllow

QuarryFrame

hookDecision? : *PermissionDecision*

needsPermission? : *ZBOOL*

sessionPhase = *spProcessing*

toolPhase = *tpPreHook*

hookDecision? = *pdAllow*

toolPhase' = **if** *needsPermission?* = *ztrue* **then** *tpPermission* **else** *tpExecuting*

sessionPhase' = *sessionPhase*

sessionId' = *sessionId*

permMode' = *permMode*

currentTool' = *currentTool*

toolsThisTurn' = *toolsThisTurn*

activeSubagents' = *activeSubagents*

context' = *context*

pendingPrompt' = *pendingPrompt*

pendingResponse' = *pendingResponse*

9.3 PreToolHookDeny

PreToolHookDeny

QuarryFrame

hookDecision? : *PermissionDecision*

denialContext? : *CONTEXTCHUNK*

sessionPhase = *spProcessing*

toolPhase = *tpPreHook*

hookDecision? = *pdDeny*

toolPhase' = *tpNone*

currentTool' = $\emptyset$

context' = $(1 \dots \text{maxContext}) \triangleleft (\text{context} \hat{\ } \langle \text{denialContext?} \rangle)$

sessionPhase' = *sessionPhase*

sessionId' = *sessionId*

permMode' = *permMode*

toolsThisTurn' = *toolsThisTurn*

activeSubagents' = *activeSubagents*

pendingPrompt' = *pendingPrompt*

pendingResponse' = *pendingResponse*

9.4 GrantPermission

GrantPermission

QuarryFrame

sessionPhase = *spProcessing*

toolPhase = *tpPermission*

toolPhase' = *tpExecuting*

sessionPhase' = *sessionPhase*

sessionId' = *sessionId*

permMode' = *permMode*

currentTool' = *currentTool*

toolsThisTurn' = *toolsThisTurn*

activeSubagents' = *activeSubagents*

context' = *context*

pendingPrompt' = *pendingPrompt*

pendingResponse' = *pendingResponse*

9.5 DenyPermission

DenyPermission
QuarryFrame
denialContext? : *CONTEXTCHUNK*

sessionPhase = *spProcessing*
toolPhase = *tpPermission*
toolPhase' = *tpNone*
currentTool' = $\emptyset$
context' = $(1 \dots \text{maxContext}) \triangleleft (\text{context} \hat{\ } \langle \text{denialContext?} \rangle)$
sessionPhase' = *sessionPhase*
sessionId' = *sessionId*
permMode' = *permMode*
toolsThisTurn' = *toolsThisTurn*
activeSubagents' = *activeSubagents*
pendingPrompt' = *pendingPrompt*
pendingResponse' = *pendingResponse*

9.6 CompleteToolSuccess

CompleteToolSuccess
QuarryFrame
output? : *CONTEXTCHUNK*

sessionPhase = *spProcessing*
toolPhase = *tpExecuting*
toolPhase' = *tpNone*
currentTool' = $\emptyset$
toolsThisTurn' = *toolsThisTurn* + 1
context' = $(1 \dots \text{maxContext}) \triangleleft (\text{context} \hat{\ } \langle \text{output?} \rangle)$
sessionPhase' = *sessionPhase*
sessionId' = *sessionId*
permMode' = *permMode*
activeSubagents' = *activeSubagents*
pendingPrompt' = *pendingPrompt*
pendingResponse' = *pendingResponse*

9.7 CompleteToolFailure

CompleteToolFailure
QuarryFrame
errorContext? : *CONTEXTCHUNK*

sessionPhase = *spProcessing*
toolPhase = *tpExecuting*
toolPhase' = *tpNone*
currentTool' = $\emptyset$
toolsThisTurn' = *toolsThisTurn* + 1
context' = $(1 \dots \text{maxContext}) \triangleleft (\text{context} \hat{\ } \langle \text{errorContext?} \rangle)$
sessionPhase' = *sessionPhase*
sessionId' = *sessionId*
permMode' = *permMode*
activeSubagents' = *activeSubagents*
pendingPrompt' = *pendingPrompt*
pendingResponse' = *pendingResponse*

10 Quarry Operations

10.1 CaptureWebFetch

PostToolUse on **WebFetch**. When web fetch capture is enabled, the fetched URL is added to the captured set and ingested into the **web-captures** collection (async, fire-and-forget).

CaptureWebFetch

$\Delta State$

url? : *URL*

```
sessionPhase = spProcessing
toolPhase = tpNone
cfgWebFetch = ztrue
url?  $\notin$  capturedUrls
#capturedUrls < maxCaptures
capturedUrls' = capturedUrls  $\cup$  {url?}
sessionPhase' = sessionPhase
sessionId' = sessionId
permMode' = permMode
toolPhase' = toolPhase
currentTool' = currentTool
toolsThisTurn' = toolsThisTurn
activeSubagents' = activeSubagents
context' = context
pendingPrompt' = pendingPrompt
pendingResponse' = pendingResponse
codebaseIndexed' = codebaseIndexed
activeDb' = activeDb
compactionSaved' = compactionSaved
cfgSessionSync' = cfgSessionSync
cfgWebFetch' = cfgWebFetch
cfgCompaction' = cfgCompaction
```

10.2 SwitchDatabase

Claude calls **quarry use <name>**. Changes the active database for subsequent queries.

SwitchDatabase

$\Delta State$

db? : *DBNAME*

sessionPhase = *spProcessing*
toolPhase = *tpNone*
activeDb' = {*db?*}
sessionPhase' = *sessionPhase*
sessionId' = *sessionId*
permMode' = *permMode*
toolPhase' = *toolPhase*
currentTool' = *currentTool*
toolsThisTurn' = *toolsThisTurn*
activeSubagents' = *activeSubagents*
context' = *context*
pendingPrompt' = *pendingPrompt*
pendingResponse' = *pendingResponse*
codebaseIndexed' = *codebaseIndexed*
capturedUrls' = *capturedUrls*
compactionSaved' = *compactionSaved*
cfgSessionSync' = *cfgSessionSync*
cfgWebFetch' = *cfgWebFetch*
cfgCompaction' = *cfgCompaction*

10.3 SearchKnowledge

Claude calls `quarry find`. This is a read-only query that injects search results into context. Modeled to show that search results flow into the conversation.

SearchKnowledge

$\Delta State$

results? : *CONTEXTCHUNK*

sessionPhase = *spProcessing*
toolPhase = *tpNone*
context' = $(1 \dots \text{maxContext}) \triangleleft (\text{context} \hat{\cup} \langle \text{results?} \rangle)$
sessionPhase' = *sessionPhase*
sessionId' = *sessionId*
permMode' = *permMode*
toolPhase' = *toolPhase*
currentTool' = *currentTool*
toolsThisTurn' = *toolsThisTurn*
activeSubagents' = *activeSubagents*
pendingPrompt' = *pendingPrompt*
pendingResponse' = *pendingResponse*
codebaseIndexed' = *codebaseIndexed*
activeDb' = *activeDb*
capturedUrls' = *capturedUrls*
compactionSaved' = *compactionSaved*
cfgSessionSync' = *cfgSessionSync*
cfgWebFetch' = *cfgWebFetch*
cfgCompaction' = *cfgCompaction*

10.4 InjectKnowledgeHint

Quarry proactively injects relevant knowledge into context at key moments: **SessionStart** (project standards, workflow tips), **PostToolUse** on **Edit|Write** (e.g., “remember our changelog standard”, “update DESIGN.md with an ADR”), **PostToolUse** on tool-specific patterns (e.g., “top hints for working with Z specs” when editing `.tex` files).

These are contextual hints retrieved from quarry collections (standards, design docs, project conventions) and injected as **additionalContext**. The injection is non-blocking and additive — it appends to context without altering other state.

<i>InjectKnowledgeHint</i>	
$\Delta State$	
<i>hint?</i> : <i>CONTEXTCHUNK</i>	
<i>sessionPhase</i> $\in \{spIdle, spProcessing\}$	
<i>context'</i> = $(1 \dots maxContext) \triangleleft (context \hat{\ } \langle hint? \rangle)$	
<i>sessionPhase'</i> = <i>sessionPhase</i>	
<i>sessionId'</i> = <i>sessionId</i>	
<i>permMode'</i> = <i>permMode</i>	
<i>toolPhase'</i> = <i>toolPhase</i>	
<i>currentTool'</i> = <i>currentTool</i>	
<i>toolsThisTurn'</i> = <i>toolsThisTurn</i>	
<i>activeSubagents'</i> = <i>activeSubagents</i>	
<i>pendingPrompt'</i> = <i>pendingPrompt</i>	
<i>pendingResponse'</i> = <i>pendingResponse</i>	
<i>codebaseIndexed'</i> = <i>codebaseIndexed</i>	
<i>activeDb'</i> = <i>activeDb</i>	
<i>capturedUrls'</i> = <i>capturedUrls</i>	
<i>compactionSaved'</i> = <i>compactionSaved</i>	
<i>cfgSessionSync'</i> = <i>cfgSessionSync</i>	
<i>cfgWebFetch'</i> = <i>cfgWebFetch</i>	
<i>cfgCompaction'</i> = <i>cfgCompaction</i>	

This operation fires in both **spIdle** (SessionStart context injection) and **spProcessing** (PostToolUse context injection). The hint content is determined by quarry’s semantic search over registered collections — the Z spec abstracts this as an opaque **CONTEXTCHUNK** input.

Example injection points:

- **SessionStart**: “This project follows punt-kit standards. Key conventions: ruff, mypy, pytest, CHANGELOG.md under [Unreleased].”
- **PostToolUse on Edit of CHANGELOG.md**: “Keep a Changelog format. Entries under [Unreleased] with Added/Changed/Fixed sections.”
- **PostToolUse on Edit of *.tex**: “Z spec conventions: use `\quad` for indentation, ZBOOL for booleans, bounded sets for ProB animation.”
- **PostToolUse on Bash** with git commands: “Session close protocol: git status, git add, bd sync, git commit, git push.”

11 Subagent Operations

SpawnSubagent

QuarryFrame

agentId? : *AGENTID*

sessionPhase = *spProcessing*
agentId? $\notin$ *activeSubagents*
#activeSubagents < *maxSubagents*
activeSubagents' = *activeSubagents* $\cup$ {*agentId?*}
sessionPhase' = *sessionPhase*
sessionId' = *sessionId*
permMode' = *permMode*
toolPhase' = *toolPhase*
currentTool' = *currentTool*
toolsThisTurn' = *toolsThisTurn*
context' = *context*
pendingPrompt' = *pendingPrompt*
pendingResponse' = *pendingResponse*

StopSubagent

QuarryFrame

agentId? : *AGENTID*

hookResult? : *HookDecision*

resultContext? : *CONTEXTCHUNK*

sessionPhase = *spProcessing*
agentId? $\in$ *activeSubagents*
hookResult? = *hdAllow*
activeSubagents' = *activeSubagents* $\setminus$ {*agentId?*}
context' = (1 .. *maxContext*) $\triangleleft$ (*context* $\cap$ {*resultContext?*})
sessionPhase' = *sessionPhase*
sessionId' = *sessionId*
permMode' = *permMode*
toolPhase' = *toolPhase*
currentTool' = *currentTool*
toolsThisTurn' = *toolsThisTurn*
pendingPrompt' = *pendingPrompt*
pendingResponse' = *pendingResponse*

12 Response and Stop

BeginResponse

QuarryFrame

response? : *RESPONSE*

sessionPhase = *spProcessing*
toolPhase = *tpNone*
activeSubagents = $\emptyset$
sessionPhase' = *spResponding*
pendingResponse' = {*response?*}
pendingPrompt' = $\emptyset$
sessionId' = *sessionId*
permMode' = *permMode*
toolPhase' = *tpNone*
currentTool' = $\emptyset$
toolsThisTurn' = *toolsThisTurn*
activeSubagents' = *activeSubagents*
context' = *context*

FinishResponse

QuarryFrame

hookResult? : *HookDecision*

responseContext? : *CONTEXTCHUNK*

sessionPhase = *spResponding*

hookResult? = *hdAllow*

sessionPhase' = *spIdle*

pendingResponse' = $\emptyset$

context' = $(1 \dots \text{maxContext}) \triangleleft (\text{context} \hat{\ } \langle \text{responseContext?} \rangle)$

toolsThisTurn' = 0

sessionId' = *sessionId*

permMode' = *permMode*

toolPhase' = *tpNone*

currentTool' = $\emptyset$

activeSubagents' = *activeSubagents*

pendingPrompt' = $\emptyset$

ForceContinue

QuarryFrame

hookResult? : *HookDecision*

sessionPhase = *spResponding*

hookResult? = *hdBlock*

sessionPhase' = *spProcessing*

pendingResponse' = $\emptyset$

sessionId' = *sessionId*

permMode' = *permMode*

toolPhase' = *tpNone*

currentTool' = $\emptyset$

toolsThisTurn' = *toolsThisTurn*

activeSubagents' = *activeSubagents*

context' = *context*

pendingPrompt' = *pendingPrompt*

13 Context Management — Compaction Safety

This is the critical section for quarry. Context compaction destroys conversation history. The **PreCompact** hook must capture session notes *before* the context is shrunk.

13.1 CaptureBeforeCompact

PreCompact hook fires. If compaction capture is enabled, quarry reads the transcript and ingests it as session notes into the **session-notes** collection. This must happen before **CompactContext**.

<i>CaptureBeforeCompact</i> $\Delta State$

<i>sessionPhase</i> = <i>spIdle</i> <i>cfgCompaction</i> = <i>ztrue</i> <i>#context</i> > 0 <i>compactionSaved'</i> = <i>ztrue</i> <i>sessionPhase'</i> = <i>sessionPhase</i> <i>sessionId'</i> = <i>sessionId</i> <i>permMode'</i> = <i>permMode</i> <i>toolPhase'</i> = <i>toolPhase</i> <i>currentTool'</i> = <i>currentTool</i> <i>toolsThisTurn'</i> = <i>toolsThisTurn</i> <i>activeSubagents'</i> = <i>activeSubagents</i> <i>context'</i> = <i>context</i> <i>pendingPrompt'</i> = <i>pendingPrompt</i> <i>pendingResponse'</i> = <i>pendingResponse</i> <i>codebaseIndexed'</i> = <i>codebaseIndexed</i> <i>activeDb'</i> = <i>activeDb</i> <i>capturedUrls'</i> = <i>capturedUrls</i> <i>cfgSessionSync'</i> = <i>cfgSessionSync</i> <i>cfgWebFetch'</i> = <i>cfgWebFetch</i> <i>cfgCompaction'</i> = <i>cfgCompaction</i>

13.2 SkipCaptureBeforeCompact

Compaction capture is disabled. The **PreCompact** hook is a no-op. Session notes are not saved.

<i>SkipCaptureBeforeCompact</i> $\Xi State$
--

<i>sessionPhase</i> = <i>spIdle</i> <i>cfgCompaction</i> = <i>zfalse</i>

13.3 CompactContext

Context is shrunk. The **compactionSaved** flag records whether notes were captured. After compaction it resets to **zfalse** for the next potential compaction cycle.

CompactContext

Δ State

newLen? : $\mathbb{N}$

```
sessionPhase = spIdle
#context > 0
newLen? < #context
newLen? ≤ maxContext
context' = (1 .. newLen?) ◁ context
compactionSaved' = zfalse
sessionPhase' = sessionPhase
sessionId' = sessionId
permMode' = permMode
toolPhase' = toolPhase
currentTool' = currentTool
toolsThisTurn' = toolsThisTurn
activeSubagents' = activeSubagents
pendingPrompt' = pendingPrompt
pendingResponse' = pendingResponse
codebaseIndexed' = codebaseIndexed
activeDb' = activeDb
capturedUrls' = capturedUrls
cfgSessionSync' = cfgSessionSync
cfgWebFetch' = cfgWebFetch
cfgCompaction' = cfgCompaction
```

14 Session End

EndSession

Δ State

reason? : *SessionEndReason*

```
sessionPhase = spIdle
activeSubagents = ∅
sessionPhase' = spEnded
sessionId' = ∅
toolPhase' = tpNone
currentTool' = ∅
toolsThisTurn' = 0
activeSubagents' = ∅
context' = ⟨⟩
pendingPrompt' = ∅
pendingResponse' = ∅
permMode' = permMode
codebaseIndexed' = zfalse
activeDb' = ∅
capturedUrls' = ∅
compactionSaved' = zfalse
cfgSessionSync' = cfgSessionSync
cfgWebFetch' = cfgWebFetch
cfgCompaction' = cfgCompaction
```

15 System Invariants

1. **Codebase indexed when sync enabled:** if `cfgSessionSync = ztrue` and the session started via `StartSessionWithSync`, then `codebaseIndexed = ztrue` for the duration of the session.

2. **Capture config is immutable during session:** the `cfg*` flags are set at `Init` and preserved by all operations. They cannot change mid-session.
3. **WebFetch deduplication:** `CaptureWebFetch` requires `url? ∉ capturedUrls`. No URL is ingested twice in the same session.
4. **Active database consistency:** at most one database is active. `SwitchDatabase` replaces, does not accumulate.
5. **Compaction flag lifecycle:** `CaptureBeforeCompact` sets `compactionSaved` to `ztrue`. `CompactContext` resets it to `zfalse`. This tracks whether the most recent compaction was preceded by a capture.
6. **Clean session boundaries:** all quarry state (`indexed`, `activeDb`, `captures`, `compaction flag`) is cleared on `EndSession`.

Note on the compaction safety property: The Z spec does *not* enforce that capture must precede compaction — it is possible to compact without capturing (the operations are independent). This reflects reality: Claude Code fires the pre-compact hook before compaction, but the hook is non-blocking. The `compactionSaved` flag lets us *observe* whether the capture occurred, not *enforce* it. Enforcement is the hook system’s responsibility.

16 Hook Event Map

Hook	Matcher	Operation	Effect	Blk
SessionStart	*	StartSession- WithSync/NoSync	Auto-register sync	+ No
SessionStart	*	InjectKnowledgeHint	Project standards	No
PostToolUse	WebFetch	CaptureWebFetch	Ingest URL	No
PostToolUse	Edit—Write	InjectKnowledgeHint	File-type hints	No
PostToolUse	Bash	InjectKnowledgeHint	Workflow hints	No
PostToolUse	quarry__use [†]	SwitchDatabase	Change active DB	No
PostToolUse	quarry__find [†]	SearchKnowledge	Inject results	No
PostToolUse	quarry__* [†]	(suppress output)	Format display	No
PreCompact	*	CaptureBefore- Compact	Save session notes	No
SessionEnd [‡]	*	EndSession	Clear state	No

[†]Conceptual matcher; actual regex is `mcp__(plugin_quarry...)?quarry(-proxy)?__.*` (see `hooks.json`).

[‡]Not yet wired — no `SessionEnd` hook is registered.

17 Implementation Validation

This section cross-references the Z specification against the current quarry implementation: the hook config, shell scripts in the `hooks` directory, and the Python handlers.

17.1 Hook Wiring Status

All three knowledge capture handlers are implemented in `src/quarry/hooks.py` and wired via shell scripts in `hooks/hooks.json`.

Component	Hook Event	Files	Status
Auto-register + sync	SessionStart	<code>session-start.sh</code> , <code>session-sync.sh</code> , <code>hooks.py</code>	Correct
Output formatting	PostToolUse (quarry)	<code>suppress-output.sh</code>	Correct
Web fetch capture	PostToolUse (WebFetch)	<code>web-fetch.sh</code> , <code>hooks.py</code>	Correct
Compaction capture	PreCompact	<code>pre-compact.sh</code> , <code>hooks.py</code>	Correct

17.2 Not Yet Built

Z Operation	Hook Event	Effect	Effort
InjectKnowledgeHint	SessionStart	Project standards	Medium
InjectKnowledgeHint	PostToolUse Edit—Write	File-type hints	Medium
InjectKnowledgeHint	PostToolUse Bash	Workflow hints	Medium
EndSession cleanup	SessionEnd	Clear session state	Low

Knowledge hint injection is the most significant unbuilt capability. It requires a hint resolution engine that maps the current activity (tool name, file path, command) to relevant quarry collections, plus careful scoping: hints must be short (1–3 sentences) to avoid flooding context.

An earlier attempt at convention hints via a PreToolUse hook accumulator (see ADR in `DESIGN.md`) was implemented and later removed (PR #113) in favor of a future semantic approach.

17.3 Coverage Summary

Z Operation	Implementation	Status
StartSessionWithSync	hooks.py + session-sync.sh	Correct
StartSessionNoSync	hooks.py (config check)	Correct
SubmitPrompt / RejectPrompt	(base)	Correct
Tool call operations	(base)	Correct
CaptureWebFetch	hooks.py + web-fetch.sh	Correct
SwitchDatabase	MCP tool	Correct
SearchKnowledge	MCP tool	Correct
InjectKnowledgeHint	—	Not built
CaptureBeforeCompact	hooks.py + pre-compact.sh	Correct
CompactContext	(base)	Correct
EndSession	—	Not built